A Paint-On Liquid Can Stop Cavities

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A cavity does not have to mean drilling. A simple liquid painted on a tooth in seconds can stop tooth **decay** in young children. A big test has now shown that it works.

The liquid is called silver **diamine fluoride**, or SDF. A dentist paints it on the cavity with a tiny sponge-tipped stick. There is no drilling, no shot, and no sleep medicine.

SDF stops the **decay** from getting worse. It does not put back tooth parts that are already gone. Its silver fights the germs linked to **decay**, and its **fluoride** helps make the tooth stronger.

The test, shared in JAMA **Pediatrics**, included 830 children younger than 6 from Michigan, New York, and Iowa. Margherita Fontana, a dentistry professor at the University of Michigan and the study's lead researcher, said the treatment is very effective and safe, even for children as young as 1.

There is one downside. SDF turns the **decayed** part of the tooth dark for good. The change is mostly about looks. For many families, skipping pain and drilling matters more.

Word Treasure

decay -- Damage to a tooth that makes a hole in it.

fluoride -- A mineral that helps make teeth stronger.

diamine -- Part of the name of the liquid that is painted on teeth to stop decay.

Pediatrics -- The branch of medicine that cares for children.

Background you need

Tooth decay is damage to a tooth caused by germs that live in the mouth.

Fluoride is a mineral that helps make tooth enamel stronger and protects teeth from decay.

Dentists usually treat cavities by removing the damaged part and filling the tooth, but a paint-on liquid can stop the decay without drilling.

Break it down

WHO: A dentist, young children with cavities, and lead researcher Margherita Fontana.
WHAT: A paint-on liquid called silver diamine fluoride that stops tooth decay from getting worse.
WHERE: On children's teeth, in a test with children from Michigan, New York, and Iowa.
WHY: To stop decay without drilling, shots, or sleep medicine.

Why it matters

If you ever get a cavity, this liquid could let a dentist stop it without drilling or pain.

Test what you remember

1. What is SDF?

- (A) A drill for fixing teeth
- (B) A shot that puts a child to sleep
- (C) A kind of toothpaste
- (D) A liquid painted on a tooth to stop decay

2. How does a dentist put SDF on a tooth?

- (A) With a shot
- (B) With a tiny sponge-tipped stick
- (C) With sleep medicine
- (D) With a drill

3. What does the silver in SDF do?

- (A) It fights the germs linked to decay
- (B) It makes the tooth white
- (C) It numbs the tooth
- (D) It puts back tooth parts that are gone

4. What does SDF NOT do?

- (A) Stop decay from getting worse
- (B) Put back tooth parts that are already gone
- (C) Fight germs linked to decay
- (D) Help make the tooth stronger

5. How many children were in the test?

- (A) 830 children
- (B) 100 children
- (C) 1,000 children
- (D) 50 children

6. What is one downside of SDF?

- (A) It needs drilling
- (B) It only works on adults
- (C) It causes pain
- (D) It turns the decayed part of the tooth dark for good

Answer key (try the quiz first!): 1: D 2: B 3: A 4: B 5: A 6: D

Questions to think about

1. Margherita Fontana: She said the treatment is very effective and safe, even for children as young as 1.

2. Many families: For them, skipping pain and drilling matters more than the tooth turning dark.

MY THOUGHTS (at least 20 words):